CampusSeat - University Facility Booking System

Course: Software Engineering
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1. Business description of the system

1.1 Introduction

CampusSeat is a web-based comprehensive facility management and booking system designed specifically for university environments. It allows students, academic staff, and university administrators to efficiently browse, reserve, and manage university spaces such as study rooms, auditoriums and laboratories.

1.2 Motivation to create the software

Currently, the university relies on a fragmented system of email requests and paper-based schedules managed by individual faculty secretariats. This leads to frequent double-bookings, inefficient use of space (empty rooms that appear booked), and frustration among students and staff. CampusSeat aims to digitalize and centralize this process, providing real-time availability and an automated approval workflow to save time and optimize campus resources.

1.3 Main features

- **Search & Filter:** Users can search for available rooms based on date, time, capacity, and required equipment (e.g., projectors, computers).
- **Role-Based Booking Workflow:** Students can book small study rooms instantly, while booking large auditoriums requires an administrator's approval.
- **Dashboard & Schedule View:** A personalized dashboard showing upcoming reservations and a calendar view of room availability.
- **Notifications:** Automated email alerts for booking confirmations, cancellations, and administrative rejections.

2. Software Requirements Specification (SRS)

2.1 Functional Requirements (User Stories)

Req ID	User Story
REQ-F01 (Search)	As a User (Student/Staff), I want to filter rooms by capacity and equipment so that I can find a room that meets my specific needs.
REQ-F02 (Booking)	As a User, I want to select an available time slot and submit a reservation request so that I can secure a space for my study group or lecture.
REQ-F03 (Cancellation)	As a User, I want to cancel my upcoming reservation so that the room becomes available for others.
REQ-F04 (Approval)	As an Administrator, I want to view pending reservation requests for restricted rooms and approve or reject them so that I can control access to critical university facilities.
REQ-F05 (Auth)	As a User, I want to log in using my university credentials (SSO) to access the system securely.

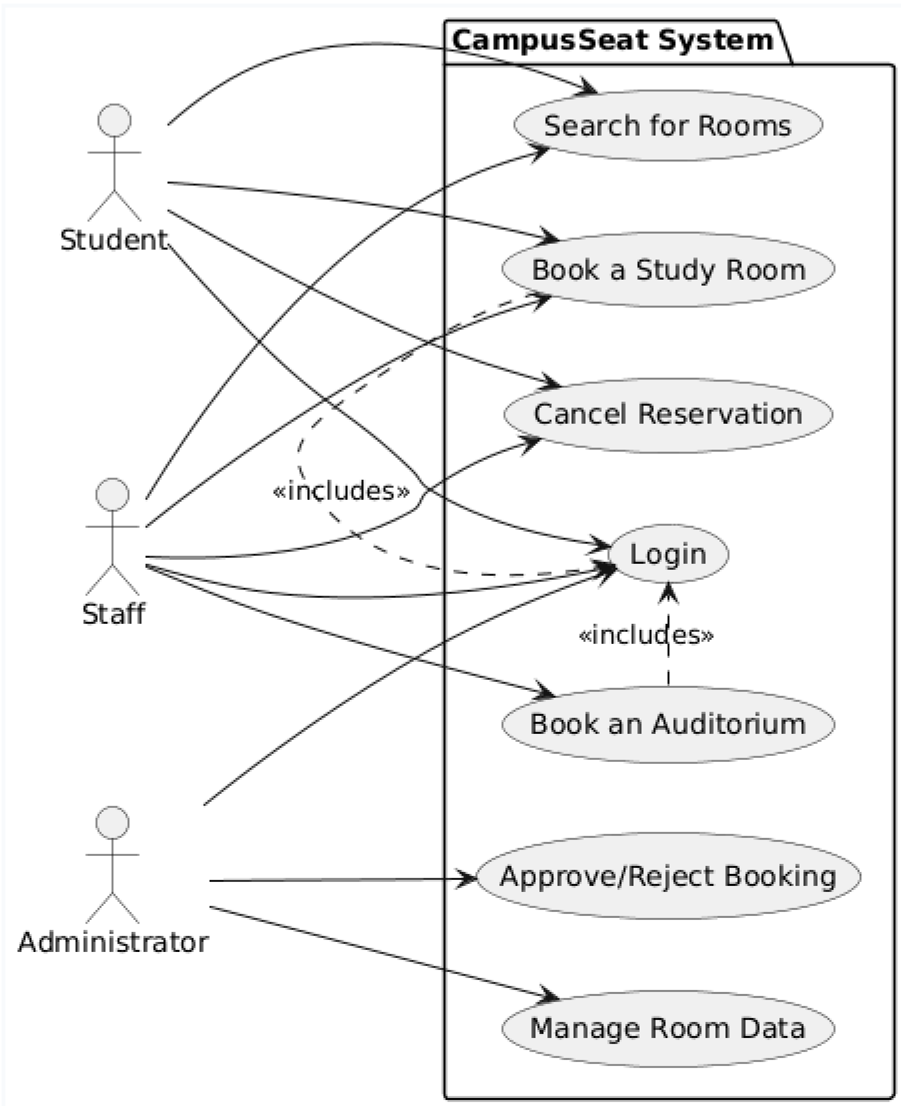
2.2 Non-Functional Requirements

Req ID	
REQ-NF01 (Performance)	The system must load the room availability calendar in under 2 seconds under normal network conditions.
REQ-NF02 (Usability)	The application interface must be responsive and fully functional on both desktop computers and mobile devices.
REQ-NF03 (Security)	All user passwords must be hashed using bcrypt, and all data transmission must be encrypted via HTTPS.
REQ-NF04 (Reliability)	The system should guarantee 99.9% uptime during the academic semester.

3. Models (Design) of projected system

3.1 Use case diagram

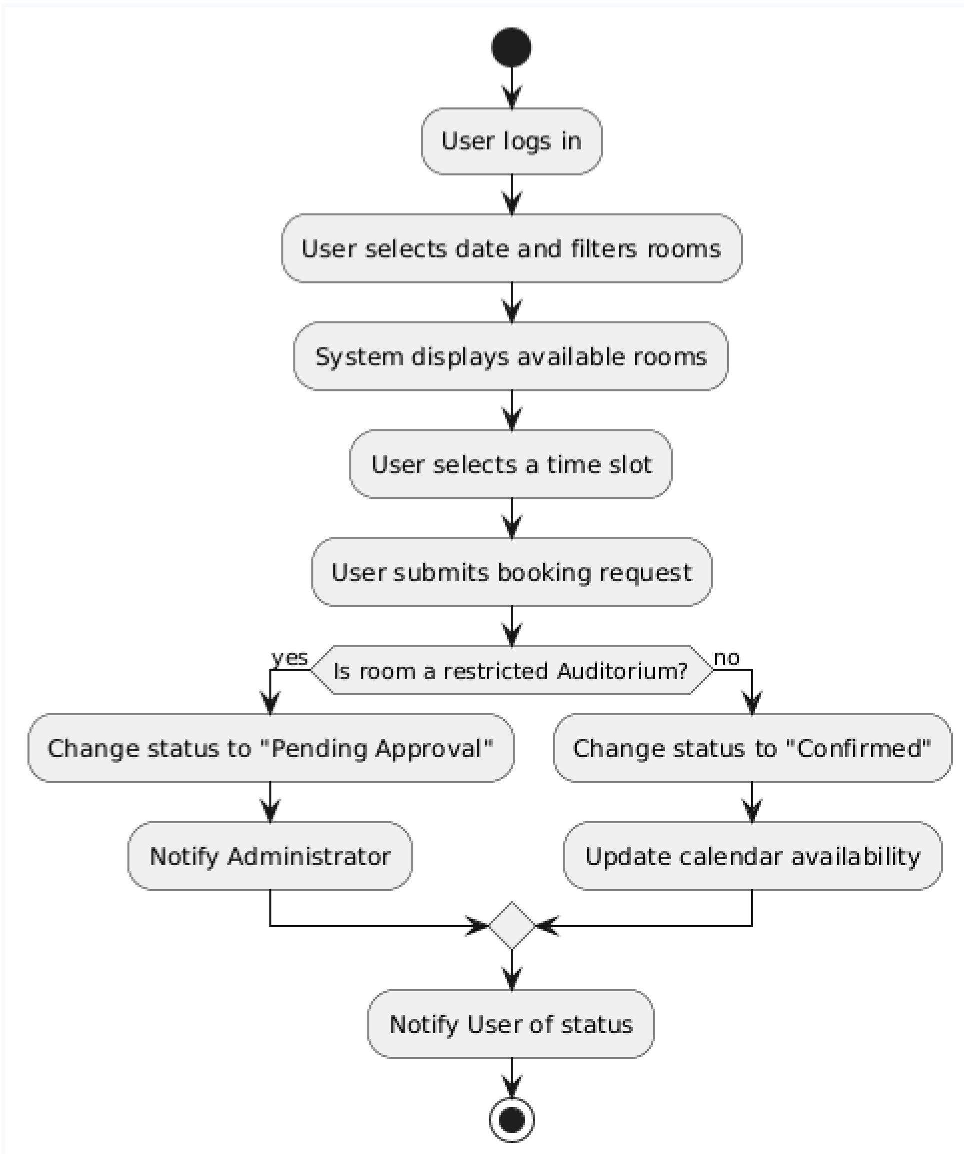
Identification of use cases and actors for the entire system.



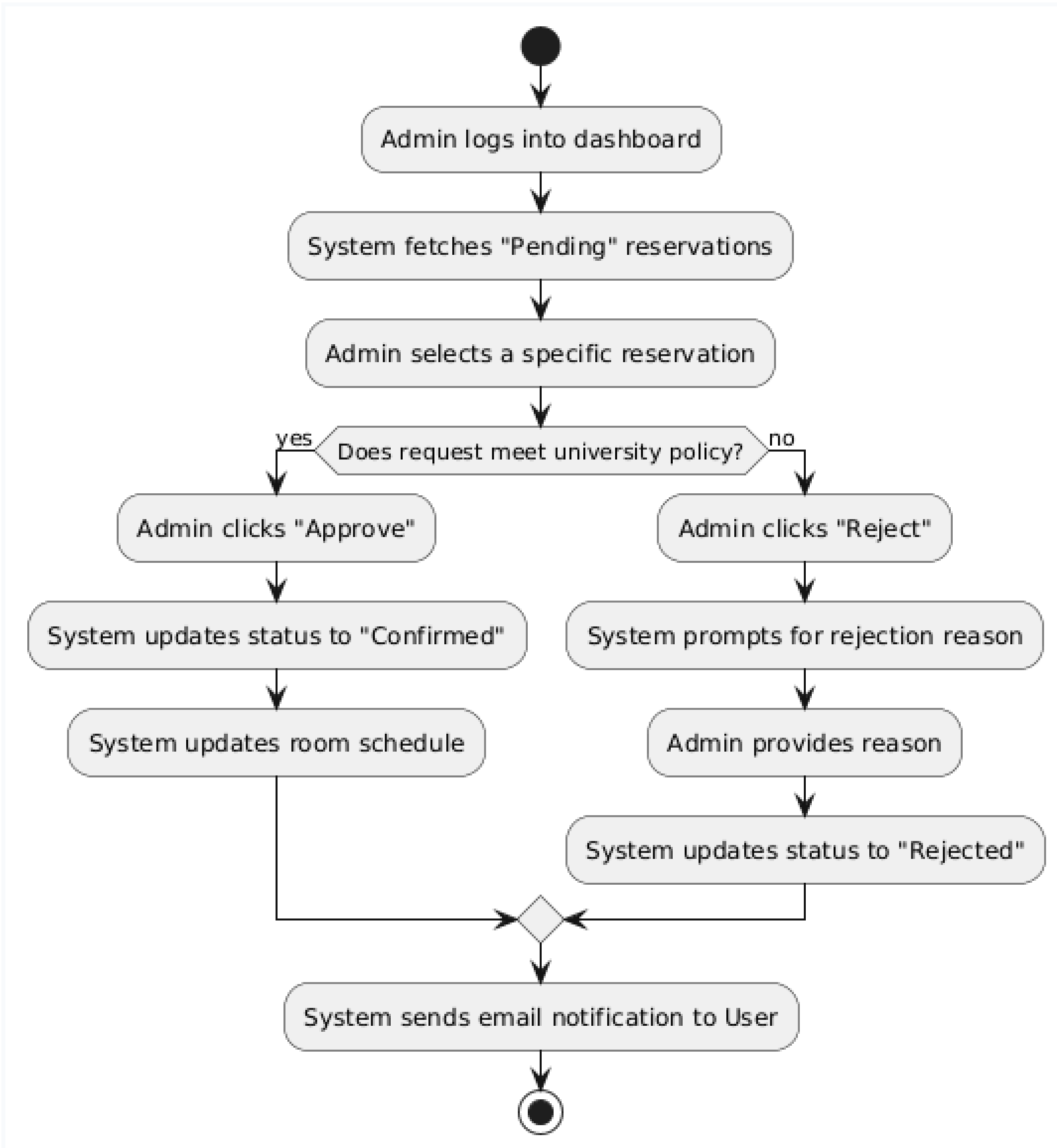
3.2 Activity diagram

2 selected aspects of the system operation in which there are decision moments.

Aspect A: Room Booking Process (Decision: Does the room require approval?)

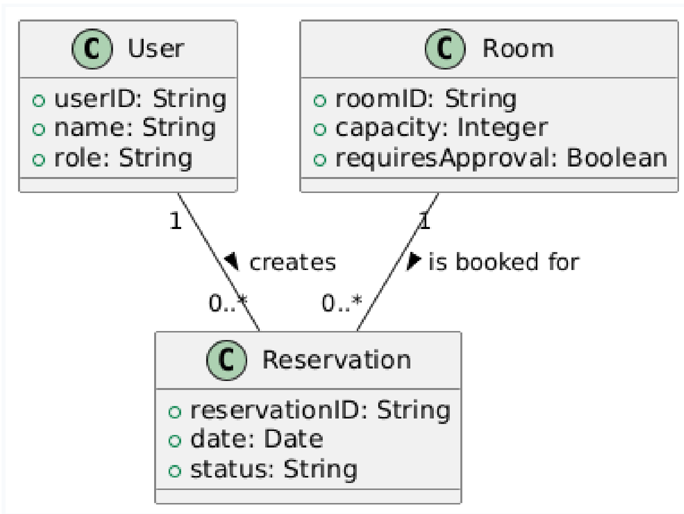


Aspect B: Admin Approval Workflow (Decision: Approve or Reject?)



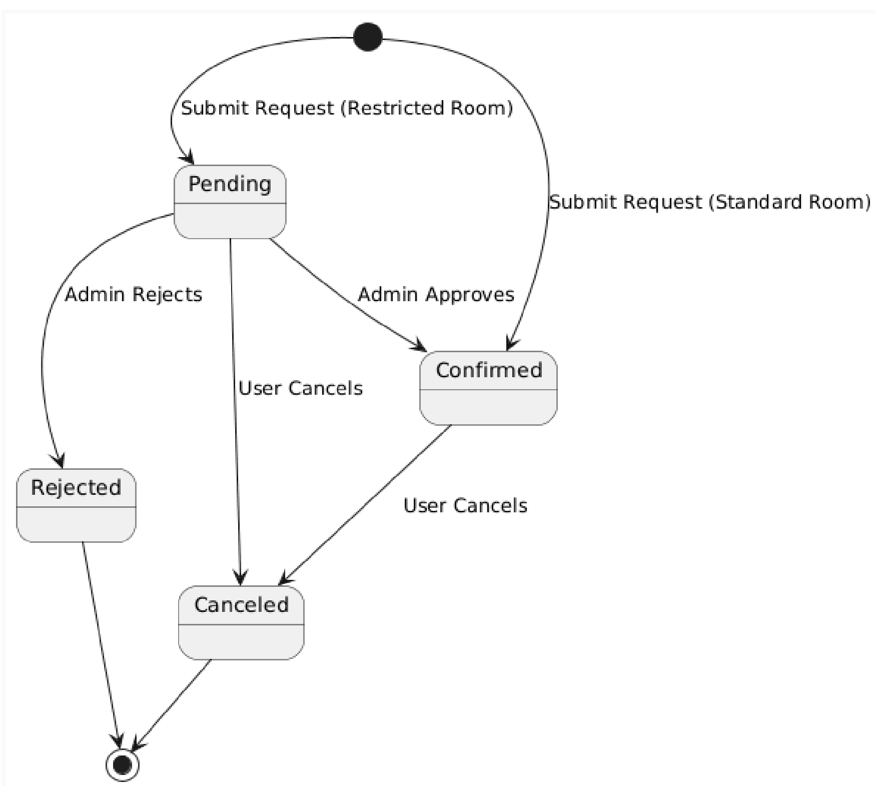
3.3 Class diagram

Domain model of entire system.



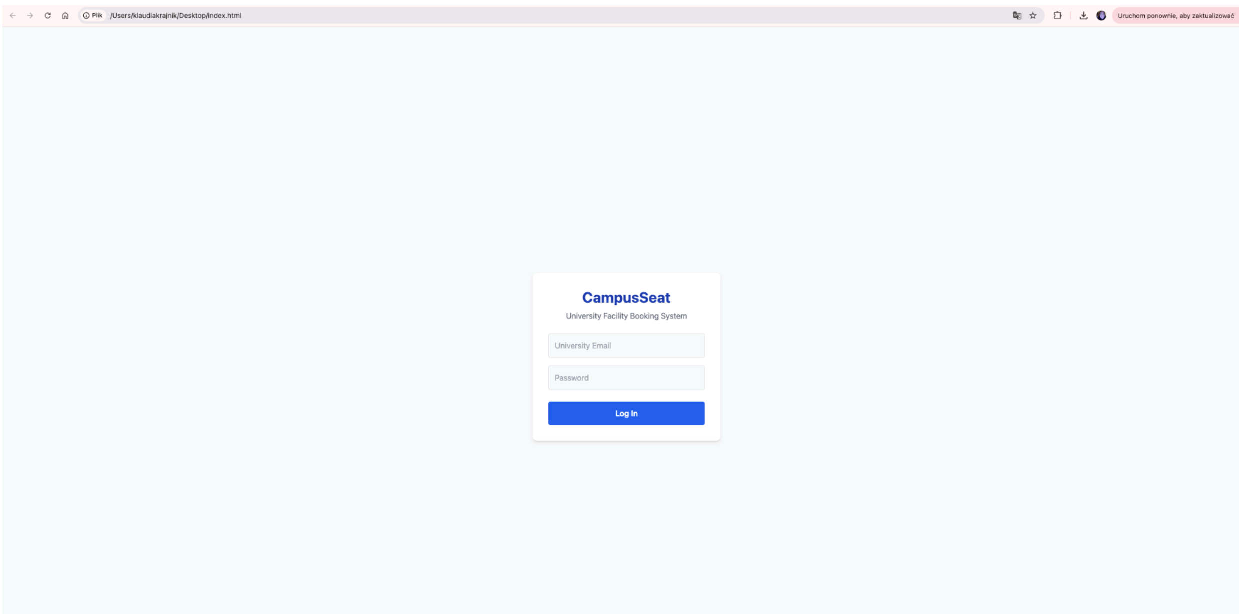
3.4 Additional diagram - State Machine

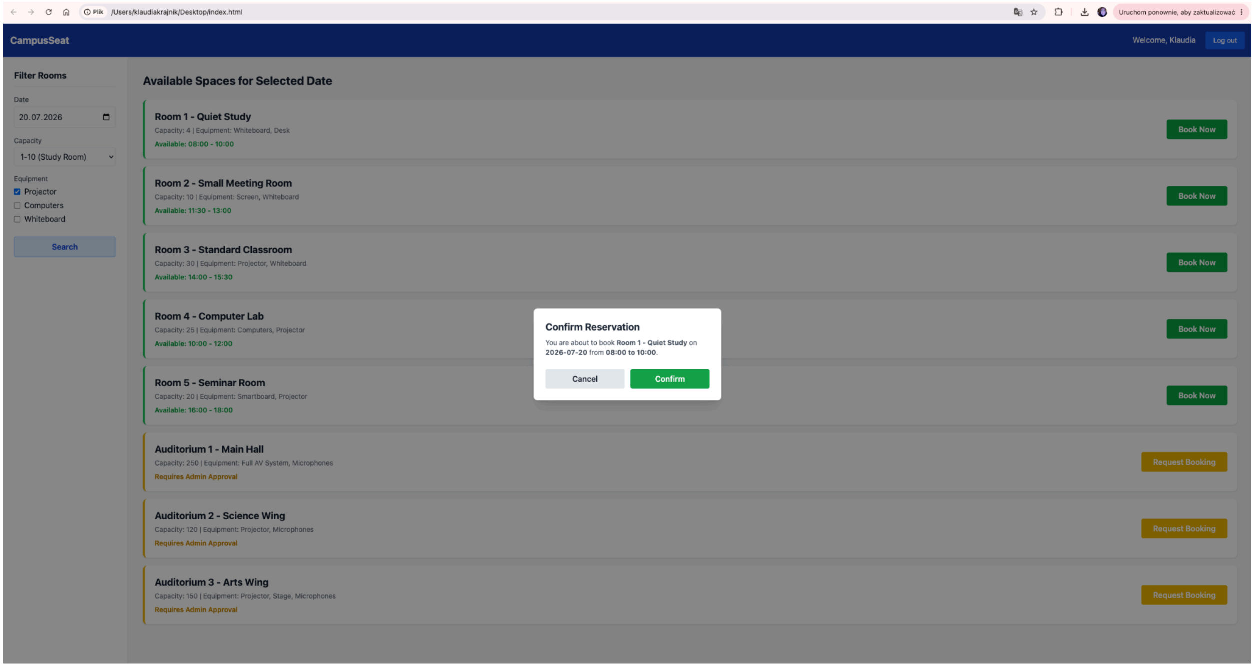
State Machine for Reservation status lifecycle.

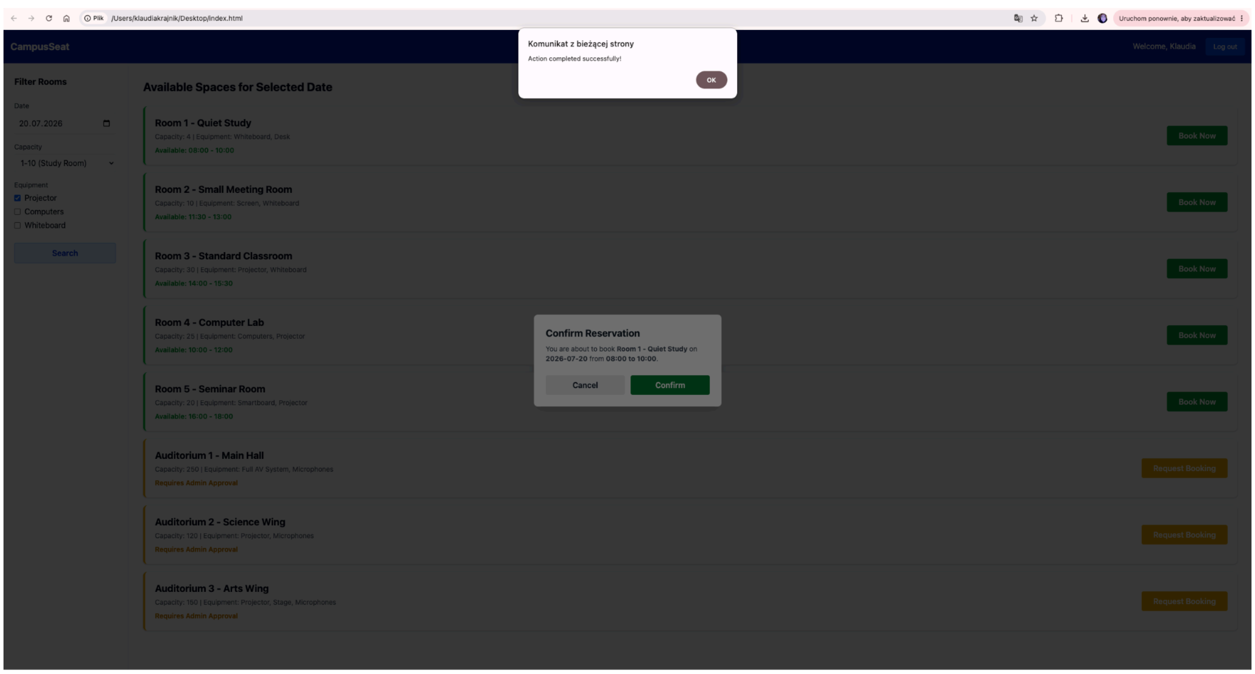


4. System Prototype

4.1 Graphic design / Mockup




/Users/klaudiakrajnik/Desktop/index.html
Uruchom ponownie, aby zaktualizować
CampusSeat
Welcome, Klaudia
Log out
Filter Rooms
Date
20.07.2026
Capacity
1-10 (Study Room)
Equipment
Projector
Computers
Whiteboard
Search
Available Spaces for Selected Date
Room 1 - Quiet Study
Capacity: 4 | Equipment: Whiteboard, Desk
Available: 08:00 - 10:00
Book Now
Room 2 - Small Meeting Room
Capacity: 10 | Equipment: Screen, Whiteboard
Available: 11:30 - 13:00
Book Now
Room 3 - Standard Classroom
Capacity: 30 | Equipment: Projector, Whiteboard
Available: 14:00 - 15:30
Book Now
Room 4 - Computer Lab
Capacity: 25 | Equipment: Computers, Projector
Available: 10:00 - 12:00
Book Now
Confirm Reservation
You are about to book Room 1 - Quiet Study on 2026-07-20 from 08:00 to 10:00.
Cancel
Confirm
Room 5 - Seminar Room
Capacity: 20 | Equipment: Smartboard, Projector
Available: 16:00 - 18:00
Book Now
Auditorium 1 - Main Hall
Capacity: 250 | Equipment: Full AV System, Microphones
Requires Admin Approval
Request Booking
Auditorium 2 - Science Wing
Capacity: 120 | Equipment: Projector, Microphones
Requires Admin Approval
Request Booking
Auditorium 3 - Arts Wing
Capacity: 150 | Equipment: Projector, Stage, Microphones
Requires Admin Approval
Request Booking


/Users/klaudiakrajnik/Desktop/index.html
Uruchom ponownie, aby zaktualizować
Komunikat z bieżącej strony
Action completed successfully!
OK
CampusSeat
Filter Rooms
Available Spaces for Selected Date
Room 1 - Quiet Study
Room 2 - Small Meeting Room
Room 3 - Standard Classroom
Room 4 - Computer Lab
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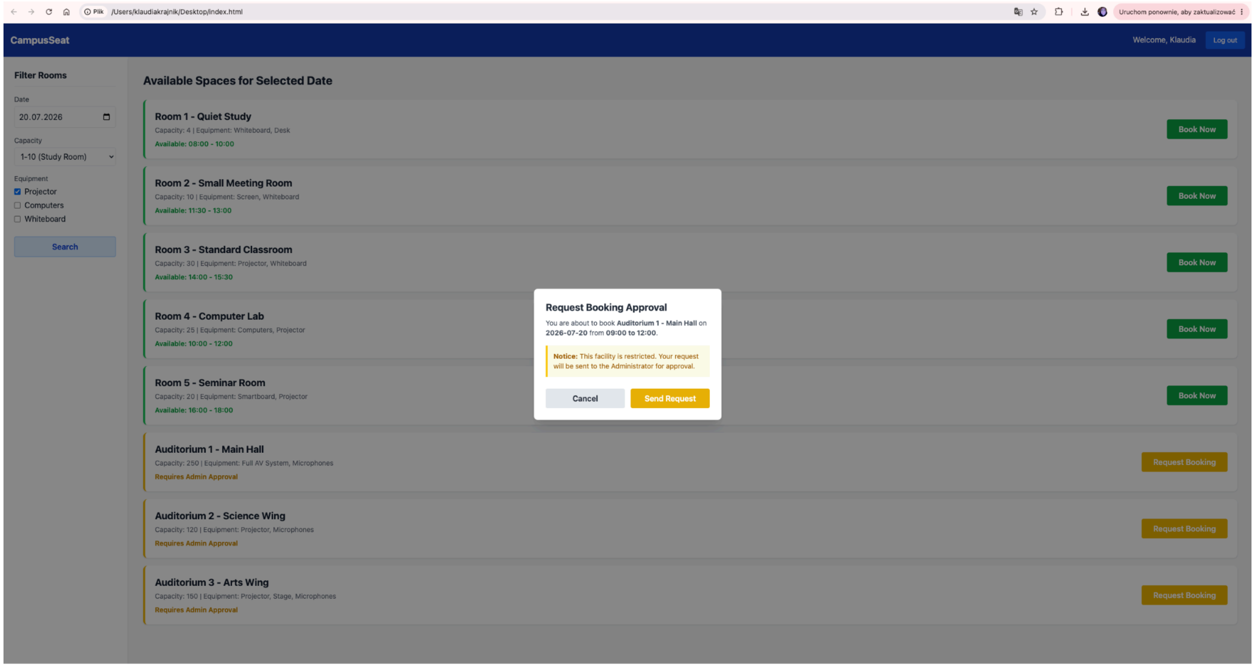
Auditorium 1 - Main Hall
Auditorium 2 - Science Wing
Auditorium 3 - Arts Wing



4.2 Implementation & Technology used

- **Programming Language:** HTML5, Vanilla JavaScript.
- **Architecture:** Single-page prototype layout with dynamic DOM manipulation for state changes (modal visibility)**.**
- **Libraries & Frameworks:** Tailwind CSS (via CDN) was strictly used for rapid, responsive user interface styling and layout structuring.
- **Repository:** Source code is maintained in a GitHub repository. The exported repository archive is provided as an attached ZIP file.